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- **In 250 words or less, explain how the nominee exemplifies each of the traits identified in [the rubric](#).**
- Aligned: Driven by an Entrepreneurial Mindset

During his career as a professor at Lawrence Tech, Eric has made connections with his research expertise in injury biomechanics and his biomedical engineering students by implementing EML teaching modules across his courses. He is dedicated to creating value in sports injury prevention by developing wearable technology devices, which could have a disruptive impact for young athletes' health and safety. He uses these real world themes to motivate problem based learning modules, starting with the EML mindset, then incorporating engineering design skills and finally communicating solutions. One of the first such modules he implemented was a collaboration with Santa Clara University and Nike Inc. to get freshman in his Introduction to Biomedical Engineering course to identify opportunities for an in-shoe pressure sensor system. Later he combined technical biomechanics concepts with EML by inviting the LTU Softball coach to interact with his class to build swing sensors to help players improve their batting performance. A final example to demonstrate Eric's continued commitment to KEEN's mission is his development of an entire course focused on the EML mindset, "Wearable Technology Design Studio". Their goal was to create a, junior-level, discipline-specific course to be part of the four-year engineering design sequence. The

wearable technology projects developed during this course have been continued by students after the course into senior projects, research projects and as business incubator projects. There have also been unique and highly creative interactions between engineers, art and design students and computer science students that have arisen from the course.

- Innovative: Pursuing innovative opportunities that create value for others

Eric has stood out from his peers with his enthusiasm, exceedingly creative ideas and great esteem he has received from the other faculty and students that have participated with the modules that he has implemented. Early on, he understood that entrepreneurship and innovation were essential to develop students who are better at adapting to new trends, embrace creativity and leadership, and understand engineering impacts on society and business. After completing initiation workshops, he truly embraced the KEEN philosophies and since then has consistently come up with new ways to incorporate those principles throughout the Biomedical Engineering program. When the Engineering College at LTU updated its multi-disciplinary engineering design curriculum he was an important part of the planning and led the development of the final design project in the foundational course. This was an open-ended and real-world PBL module that required students to use all of the skills they had been practicing. More than six years later (impacting well over 1000 students), this project continues to be one of the two key challenges that students complete. Eric has devoted his professional service to KEEN and to promoting LTU as a leader in the network for developing student focused Entrepreneurial Minded Learning and training faculty to implement similar pedagogy in their programs.

- Impact: Improving the hearts and minds of our students, peers, and communities

With more than 10 cards published on engineeringunleashed.com and six conference papers, Eric has a strong scholarship record for his KEEN related projects. He has also presented his materials nationally and internationally, not only in Engineering Education circles, but also in his disciplines in Biomedical Engineering. He used KEEN Topical Grants to develop new materials and also to share his materials and train faculty at LTU and across the network. Eric consistently engages with KEEN at their conferences, online and in subnets. He has hosted workshops at KEEN schools and other conferences, led webinars and many conference presentations. Eric continues to develop EML modules for his courses, particularly related to Biomechanics. He works with other faculty to use those modules and then publishes the technique and assessment results. Eric promotes EML at LTU and his KEEN related work was an important factor in his promotion and tenure to an Associate Professor. In addition, Eric offers similar programs for middle and high school students and advises many real world, quantified self/wearable themed, EML projects for undergraduate and graduate students.

- Inspiring: Helping others replicate and scale ideas worth spreading

The QS4EML Roadshow was a KEEN Subnet sponsored workshop that engaged at least five KEEN schools' faculty and students with representatives from industry. Faculty in this subnet had access to EML resources and developed related course modules for their programs. Many of these modules were shared between the universities and some of the collaborations resulted in conference papers or other

publications. Eric also strongly advocates for EML with his Biomedical Engineering colleagues. He has worked closely with many faculty across multiple disciplines, with a notable example that the Wearable Technology course included six guest instructors from across all four Colleges at LTU. The Industrial Design programs also adopted that course so that their students and Biomedical Engineering students would experience working on cross-disciplinary teams.

- Enduring: Ensuring future generations have access and opportunities

Eric continues to share these entrepreneurial-minded projects at the workshops he has hosted, conferences, as well as on the KEEN EngineeringUnleashed.com website. His courses are popular with students and he continues to update and add new materials and modules related to EML. Colleagues at LTU and across the network are inspired by his enthusiasm and accomplishments. Currently, Eric is leading the development of a Wearable Technology Innovation Center that will bring design, research, engineering and business experts together to work on exciting and highly disruptive products. An example of one of these projects is wearable sensors to train athletes with correct jump landing techniques that prevent knee injuries in sports. Based on Eric's history of success in biomechanics research and engineering education, it is clear that this new initiative will leave a lasting impact on Lawrence Tech, KEEN, and even global sports medicine. In summary, Eric has made a significant and positive difference at Lawrence Tech and in the profession of engineering, and I believe that he will continue to do so. He is highly deserving of the KEEN Rising Star Award.

- Provide links to relevant EM-related materials

Share links to relevant cards on EngineeringUnleashed.com and any conference/event presentation or paper, journal article, news article, etc. Please separate links by commas.

[Sensing Angular Kinematics by Embedding an Open Hardware Design Project into a Required Biomechanics Course](https://peer.asee.org/sensing-angular-kinematics-by-embedding-an-open-source-electronics-design-project-into-a-required-biomechanics-course.pdf). Meyer EG, Ulrey BL. ASEE Annual Conf Proc. 2016.

<https://peer.asee.org/sensing-angular-kinematics-by-embedding-an-open-source-electronics-design-project-into-a-required-biomechanics-course.pdf>

[Using Quantified Self as a Learning Tool to Engage Students in Entrepreneurially Minded Learning and Engineering Design](https://www.asee.org/public/conferences/64/papers/14955/download). Rust M, Nasir M, Meyer EG. ASEE Annual Conf Proc. 2016.

<https://www.asee.org/public/conferences/64/papers/14955/download>

[Project-based Learning in a Forensic Engineering Course](https://pdfs.semanticscholar.org/c037/0719283a0bc04193316b9b6a495f45c40b29.pdf). Nasir M, Weaver BT, Meyer EG. ASEE Annual Conf Proc. 2016.

<https://pdfs.semanticscholar.org/c037/0719283a0bc04193316b9b6a495f45c40b29.pdf>

[Fostering the entrepreneurial mindset through the development of multidisciplinary learning modules based on the "Quantified Self" social movement](https://www.semanticscholar.org/paper/Fostering-the-Entrepreneurial-Mindset-Through-the-Meyer-Nasir/c187144fa75ceb5943f035354de6e230664793df). Meyer EG, Nasir M. ASEE Annual Conf Proc. 2015.

<https://www.semanticscholar.org/paper/Fostering-the-Entrepreneurial-Mindset-Through-the-Meyer-Nasir/c187144fa75ceb5943f035354de6e230664793df>

[Introducing high school students to biomedical engineering through summer camps](https://peer.asee.org/20701). Nasir M, Seta J, Meyer EG. ASEE Annual Conf Proc 2014.

<https://peer.asee.org/20701>

<https://form.jotform.com/92688750547977>

[Combining discipline-specific “Introduction to Engineering” courses into a single multi-discipline course to foster the entrepreneurial mindset with entrepreneurially minded learning.](#) Gerhart A,

Carpenter D, Fletcher R, Meyer EG. ASEE Annual Conf Proc 2014

<https://peer.asee.org/20179>

KEEN’zine Issue Two, 10/2014

- Article: A Healthy Approach: A course update to encompass KEEN outcomes transforms a biomedical engineering program.

[Lawrence Technological University QS4EML Roadshow 2014](#) (Southfield, MI)

<http://blogs.ltu.edu/biomedeng/index.php/2015/05/20/quantified-self-qs-roadshow-ltu-edition/>

[Innovating the Curriculum with Entrepreneurship in Biomedical Engineering 2015](#) (Southfield, MI)

<http://blogs.ltu.edu/biomedeng/index.php/2015/04/10/bme-innovating-curriculum-with-entrepreneurial-mindset-ice-workshop-03272015/>

[Bucknell University QS4EML Roadshow 2015](#) (Lewisburg, PA)

<http://blogs.ltu.edu/biomedeng/index.php/2015/11/24/quantified-self-qs-roadshow-bucknell-edition/>

[Ohio Northern University QS4EML Roadshow 2016](#) (Ada, OH)

<http://blogs.ltu.edu/biomedeng/index.php/2016/03/15/onu-quantified-self-conference/>

American Society of Engineering Education 2017 (Columbus, OH)

- Hands-on workshop titled “[Hands-on activities for Prototyping and Ideation in Biomedical Engineering](#)” in the BME division.

<http://blogs.ltu.edu/biomedeng/index.php/2017/07/12/ltu-faculty-conduct-hands-on-workshop-at-asee-annual-conference/>

EngineeringUnleashed.com cards

<https://engineeringunleashed.com/cards/cardview.aspx?CardGuid=0af1523d-f63a-4bcc-977a-b389fa4b65b6>

<https://engineeringunleashed.com/cards/cardview.aspx?CardGuid=5fd7b863-e73d-4ad8-b688-43ace7c5c933>

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<https://engineeringunleashed.com/cards/cardview.aspx?CardGuid=8a0f3301-6557-44c2-bcea-33d5f910a6fb>